

Short Note

Seasonal altitudinal migration of White-chested Alethe *Chamaeotylas fuelleborni* in Tanzania documented with a barometric pressure logger

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Seasonal altitudinal movements from breeding areas in montane forests to non-breeding or wintering areas at lower elevations, during the cool dry season from April–June to September–October, has long been suspected for several species of birds breeding in Tanzania's Eastern Arc Mountains. This study provides new insight into the altitudinal migration of a White-chested Alethe *Chamaeotylas fuelleborni*, which was deployed with a barometric pressure logger during the cool dry season at 350 m altitude in Kimboza Forest, a lowland forest at the base of Uluguru Mountains in eastern Tanzania. The bird was found to move up to 1 500 m altitude in mid-October, most likely to montane forest in the Uluguru Mountains about 15 km away, and remained at this elevation for 3½ months, a period corresponding to the presumed breeding season, before moving down to the lowlands by the end of January. Unexpectedly, it returned to 1 500 m in mid-April where it stayed for five weeks before moving down to Kimboza Forest again where it was recaptured in July. The altitudinal migration movements were mostly started at first light about half an hour before sunrise, lasted, with one exception, no more than an hour each day and were completed in two or three days. Food is often believed to be the primary driver of altitudinal migration. The timing of the two upward movements we recorded support this hypothesis, as both coincided with the beginning of the two rainy seasons in eastern Tanzania, when army ant swarms, with which the Alethe is closely associated when foraging, become more active.

Migration altitudinale saisonnière du Cossyphe à poitrine blanche *Chamaeotylas fuelleborni* en Tanzanie, documentée à l'aide d'un enregistreur de pression barométrique

Les mouvements saisonniers dans les forêts de montagne entre les zones de reproduction en haute altitude vers les zones de non-reproduction ou d'hivernage à basse altitude pendant la saison sèche (avril-juin à septembre-octobre) ont longtemps été suspectés pour plusieurs espèces d'oiseaux se reproduisant dans les montagnes de l'Arc oriental de la Tanzanie. Cette étude donne un nouvel aperçu de la migration altitudinale des oiseaux de forêts de montagne en Afrique à partir du suivi d'un Cossyphe à poitrine blanche *Chamaeotylas fuelleborni* muni d'un enregistreur de pression barométrique, capturé pendant la saison sèche dans la forêt de plaine de Kimboza à 350 m d'altitude, au pied des monts Uluguru dans l'est de la Tanzanie. L'oiseau s'est déplacé jusqu'à 1 500 m d'altitude à la mi-octobre, très probablement vers une forêt de montagne dans les monts Uluguru à environ 15 km de là, et est resté à cette altitude pendant 3½ mois, une période correspondant à la saison de reproduction présumée, avant de redescendre dans la forêt de plaine à la fin du mois de janvier. De manière inattendue, il est retourné à 1 500 m à la mi-avril (début de la petite saison des pluies), où il est resté pendant cinq semaines avant de redescendre dans la forêt de Kimboza, où il a été recapturé en juillet. Les mouvements de migration altitudinale commençaient le plus souvent environ une demi-heure avant le lever du soleil, ne duraient, à une exception près, pas plus d'une heure par jour, et s'achevaient en deux ou trois jours. La nourriture est souvent considérée comme le principal moteur de la migration altitudinale. Les deux déplacements en altitude du Cossyphe à poitrine blanche supportent cette hypothèse puisqu'ils ont coïncidé avec le début des deux saisons des pluies dans l'est de la Tanzanie, lorsque les essaims de fourmis légionnaires — auxquels le Cossyphe est étroitement associé lorsqu'il cherche de la nourriture — deviennent plus actifs.

Keywords: afromontane forests, altitudinal movements, Eastern Arc, Uluguru Mountains

Seasonal altitudinal movements from breeding areas in montane forests to non-breeding or wintering areas at lower elevations have long been suspected for a community of birds in Africa (e.g. Chapin 1932; Irwin 1957; Oatley

1966). In the 1980s, observations indicative of seasonal movements found that at least some individuals of 26 bird species breeding in highland forest in Tanzania's Eastern Arc Mountains move to lower elevations during the cool

dry season from April–June to September–October (e.g. Stuart and Jensen 1981; Stuart and Jensen 1985; Stuart et al. 1987; Burgess and Mlingwa 2000). More recently, studies in the Uluguru Mountains finally provided tangible evidence for such movements in the Eastern Arc Mountains when Little Greenbuls *Eurillas virens* and Olive Sunbirds *Cyanomitra olivacea* were recaptured during the cold and dry season at lower elevations than where they were initially captured during the hot and wet season, and other individuals were recaptured during the hot and wet season at higher elevations than where they were initially captured during the cold and dry season (Werema 2015, 2016).

Here we provide further insights into seasonal altitudinal migration in Tanzania from a White-chested Alethe *Chamaetlas fuelleborni* deployed with a barometric pressure logger. The White-chested Alethe is a fairly large thrush-like chat that is a common inhabitant of the ground stratum of mid-elevation and montane forests at about 750–2 600 m in eastern Tanzania, northern Malawi and northeastern Zambia (Britton 1980; Jensen and Brøgger-Jensen 1992; Dowsett-Lemaire 2006; Dowsett et al. 2008). It also occurs in central Mozambique, where it is mostly found in coastal lowland forest (Clancey and Lawson 1969; Davies et al. 2011). It is often strongly associated with army ant swarms when foraging in montane forests (Dowsett-Lemaire 1989), where it feeds mainly on other arthropods flushed by the ants (Willis 1985; Dowsett-Lemaire 2006). It has long been suspected that some alethes in Tanzania and northern Malawi leave the montane forests after breeding and spend the cool dry season at lower elevations, with some moving to lower elevations on the mountains where they breed, and others leaving the montane forest blocks and moving to isolated patches of lowland forest in the foothills (Jensen et al. 1985; Stuart and Jensen 1985; Dowsett-Lemaire 1989).

One of these isolated lowland forests is Kimboza Forest Reserve at the eastern base of the Uluguru Mountains (Figure 1), where considerable numbers of White-chested Alethes have been recorded in June–July during the cool dry season but not in the warm and wet time of the year (Stuart and Jensen 1985; Svendsen et al. 1995; Werema et al. 2016b). Kimboza Forest is a remnant of lowland forest that was probably once continuous with forest on the Uluguru Mountains. Today, it includes 4 km² of semi-evergreen forest with a closed canopy to 20–30 m, and with emergent trees up to 40 m (Rodgers et al. 1983). The lowest altitude of significant areas of forest in the Uluguru Mountains is at ~1 000 m elevation (Burgess et al. 2002), around 12 km from Kimboza Forest. The area in between is covered by open woodland with scattered villages surrounded by cultivated land, but there are also a few fragments of closed forest including Mangala Forest Reserve (29 ha) at 300–675 m elevation, Milawilila Forest Reserve (13 ha) at 320–400 m, and Kisewe Forest Reserve at 500–600 m (Werema et al. 2016a; Kacholi 2020) (see Figure 1). The northeastern regions of Tanzania, including the Uluguru Mountains, have a bimodal rainfall regime with ‘short rains’ during October to January and ‘long rains’ from March to May (Kimaro et al. 2008; Mbigi and Xiao 2021). The average monthly precipitation at Tegetero Mission, at 990 m elevation on the eastern slope of the Uluguru

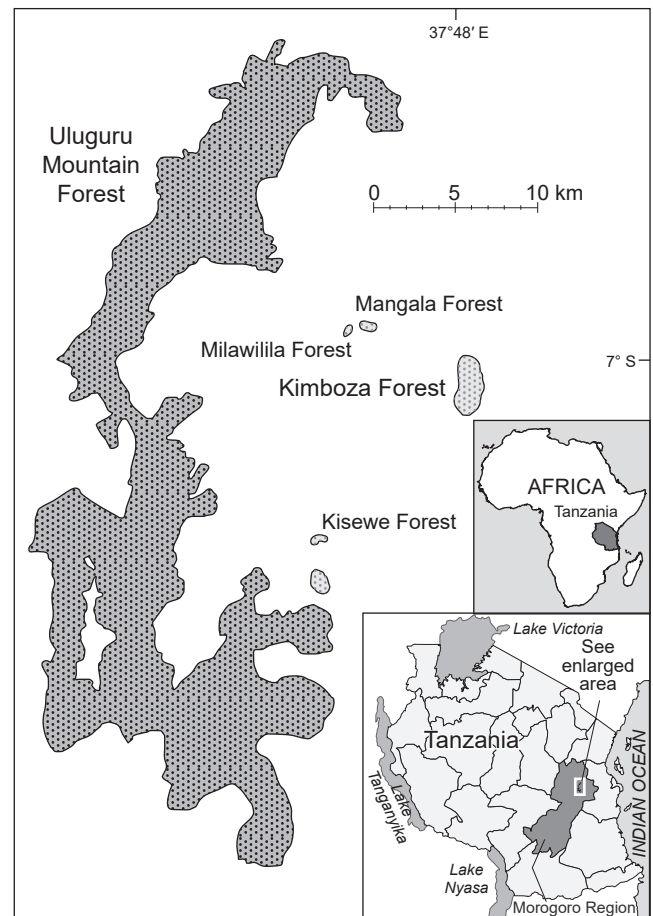


Figure 1: The distribution of highland forest in the Uluguru Mountains, Tanzania, and the location of Kimboza Forest and other lowland forest remnants at the eastern foothills

Mountains, is shown as a dotted line in Figure 1 (DHV Consulting Engineers 1980).

In Kimboza Forest we deployed barometric pressure loggers (Intigeo-CARW30Z11-DIP from Migrate Technology) on nine White-chested Alethes caught in mist nets on 22–25 August 2023. The loggers were deployed using a leg-loop harness of 1 mm nylon cord with glued knots to mount them on the back of the bird (Rappole and Tipton 1991). The weight of the logger including harness (1.2 g) represented a maximum of 2.5% of the body weight of the birds (48–61 g). The loggers stored information on barometric pressure every 20 minutes (as well as several parameters not used here) with an accuracy of ± 1 hPa corresponding to ± 10 m. By recording barometric pressure changes, it was possible to collect information on the bird's altitudinal migration, including the time spent at different altitudes.

One logger was retrieved on 24 July 2024 within 100 m of where it was deployed on the bird 11 months earlier. The collected data were first compensated for day-to-day variations in atmospheric pressure using local pressure-level data from a metrological station in Morogoro (06°50'S, 37°39'E; altitude 525 m) 25 km from Kimboza Forest, and we then converted the logger data to altitude in meters using the following version of the barometric formula (from

Shiple et al. 2017):

$$A = \frac{\left(\left(\frac{P_0}{P} \right)^{\frac{1}{5.257}} - 1 \right) (T + 273.15)}{0.0065}$$

where A = data-logger altitude (in meters), P_0 = sea-level pressure (in pascals), P = data-logger pressure (in pascals) and T = temperature at sea level (in degrees Celsius). All altitudes given here correspond to height above sea level, rounded to the nearest 10 metres; all times given are local time in Tanzania.

During the 11-month period, the bird made the following significant altitude changes:

- In mid-October, it moved from 340 m elevation up to 1 500 m where it stayed until the end of January, when it returned to the lowlands at 340 m elevation.
- In mid-April, it moved up to 1 500 m elevation again, and stayed there until the end of May, when it moved down to 350 m elevation.

The altitudinal migration movements were mostly started at first light about half an hour before sunrise and were completed in two or three days:

1. The first upward movement started at 05:20 on 15 October, when the bird moved from 340 m elevation up to 980 m in 40–60 minutes. The same afternoon, at 18:00, it continued up to 1 100 m in 20–40 minutes. The next morning, at 05:40, it reached the elevation of 1 500 m in 20–40 minutes.
2. The first downward movement was initiated at 06:00 on 25 January when the bird moved from 1 500 m altitude down to 590 m in 20 minutes at the most. The next morning, at 06:00, it continued down to 390 m elevation in 20 minutes at the most.
3. On 16 April, at 06:00, the bird moved from 330 m up to 500 m elevation in 20–40 minutes. The next morning, at 06:00, it moved up to 750 m elevation in 20–40 minutes. At 10:40 on the same day it moved on to 950 m elevation in 40–60 minutes. The following day (18 April), at 06:00, it continued up to 1 400 m elevation in 40–60 minutes. In the next days it slowly moved up to 1 500 m elevation.
4. The second downward movement started at 06:00 on 21 May when the bird moved from 1 500 m elevation down to 650 m in a maximum of 20 minutes. The next day, at 06:00, it continued down to 350 m elevation in 20 minutes at the most.

Data from the barometric pressure logger do not make it possible to determine the bird's migration route. However, by combining the time taken by the bird to reach locations at other altitudes with the topography of the surroundings, it is possible to suggest the bird's likely route. Following deployment of the data logger in late-August, the White-chested Alethe stayed at about 340 m elevation until mid-October (Table 1, Figure 2). Since Kimboza Forest is at 220–400 m elevation and surrounded by hills with open vegetation it must be assumed that this forest bird remained in Kimboza Forest throughout this period. When leaving the lowlands on 15 October, the short flying time to 1 500 m elevation, together with the two return flights to lowland forest in January and May completed in less than an hour's flying

time, strongly suggest that the bird migrated to the Uluguru Mountains, ~15 km from Kimboza, where montane forest is present at 1 500 m (Figure 2). The closest alternatives would be the montane forest on Nguru Mountains, 95 km away, or Ukaguru Mountains, 105 km away.

The breeding phenology of the White-chested Alethe is well studied on the Nyika Plateau in northern Malawi, where it peaks at the end of the dry season and beginning of the rainy season with egg-laying mostly in November (Dowsett and Dowsett-Lemaire 1984; Dowsett-Lemaire and Dowsett 2006). Breeding in the Uluguru Mountains is likely to follow the phenology of the Nyika Plateau since: the rainy season starts at about the same time throughout the species' breeding range; in the Udzungwa Mountains (100 km southwest of Uluguru Mountains) an active nest was recorded in November (Jensen et al. 1985) and fledged young in late December and mid-February (Jensen 2022; Natural History Museum of Denmark specimens). The arrival of the bird with the logger to montane forest in mid-October would coincide with the start of the breeding season. During the first 36 days, the bird spent the night at variable elevations before settling in at 1 500 m from 22 November for the next 63 days (Table 1). This could suggest that the bird started breeding in late November and spent the nights in its territory from this date. The incubation period and duration of the nestling period are unknown but if laying was started in late November, young would most probably be independent by late January, when the bird with the logger left the mountain.

Seasonal altitudinal migration in the Eastern Arc Mountains is believed to include one movement down to lower elevations at the beginning of the cool dry season, in April–June, and a return to the highland forests in September–October, with the start of the warmer and wetter season (Stuart and Jensen 1981; Stuart and Jensen 1985; Burgess and Mlingwa 2000). The early return to the lowlands at the end of January of the bird with the logger and a second, but much shorter visit to the highlands in April–May was therefore unexpected. It seems unlikely the bird returned to breed again in April, as this time of the year is outside the species' known breeding season and the stay was also too short to be able to complete a breeding cycle. Instead, this movement could be linked to food availability. Food is often believed to be the primary extrinsic driver of altitudinal migration, although little work has been dedicated to the role of insect abundance in these movements (Pageau et al. 2020). White-chested Alethe is highly dependent on army ant swarms when feeding (Willis 1985; Dowsett-Lemaire 1989), and ants are known to become irregular and local or inactive in the dry season in the montane forests in north Malawi (Dowsett-Lemaire 1987; Dowsett-Lemaire and Dowsett 2006) – and most likely also in the highlands of Tanzania. The arrival of the bird at 1 500 m elevation coincided with the start of the two rainy seasons (Figure 2), when increased ant activity and food is expected, strongly supporting that the altitudinal movement is associated with insect abundance.

We cannot explain why only one of the nine birds with a logger was recaptured. The mortality of adult White-chested Alethe was found to be low in a study in Malawi (Dowsett 1985), suggesting that most of the birds with loggers should

Table 1: Data measured at midnight during the time when a White-chested Alethe *Chamaeotylas fuelleborni* fitted with a barometric pressure logger was not making migration movements

Period	Days	Average altitude (m)	Altitudinal range (m)
2023: 24 Aug–14 Oct	52	340	25 (320–355)
2023: 17 Oct–21 Nov	36	1 449	189 (1 349–1 538)
2023–2024: 22 Nov–25 Jan	63	1 504	35 (1 488–1 523)
2024: 28 Jan–31 Jan	4	260	100 (210–310)
2024: 1 Feb–13 Apr	69	335	34 (322–356)
2024: 17 Apr–22 Apr	5	1 478	165 (1 356–1 521)
2024: 23 Apr–21 May	29	1 517	46 (1 485–1 531)
2024: 24 May–23 Jul	61	345	40 (327–367)

Note: The pressure sensor has an absolute accuracy of ± 1 hPa corresponding to ± 10 m

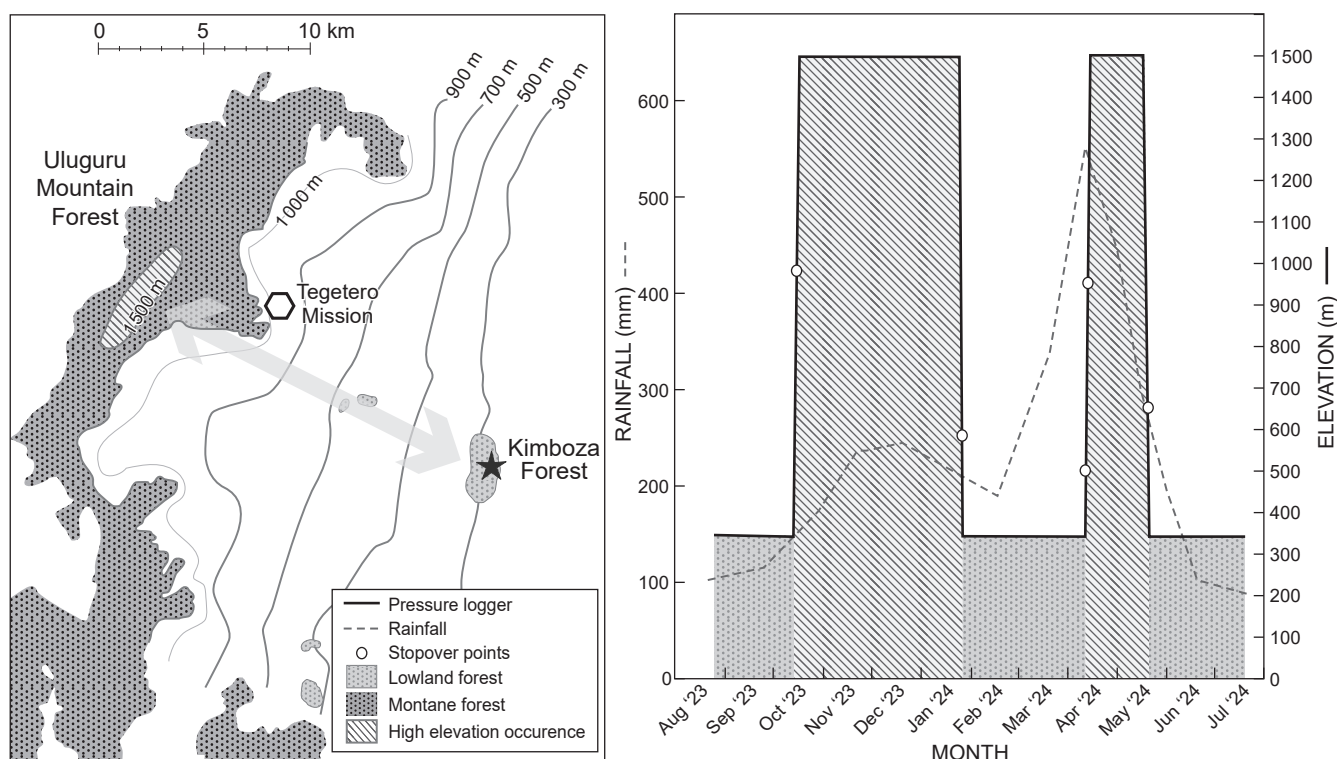


Figure 2: A altitudinal positioning of an individual White-chested Alethe *Chamaeotylas fuelleborni* fitted with a barometric pressure logger during a 11-month study period (black line), with stopovers during the migration between lowland and montane forest (small circles). The star in Kimboza Forest indicates where the Alethe was deployed with the logger and later recaptured. The hatched area in the Uluguru Mountains indicates the likely position of the Alethe when at 1 500 m elevation, and the arrow between this area and Kimboza Forest indicates the likely migration route. The dotted line indicates the average monthly precipitation at Tegetero Mission, at 990 m elevation on the eastern slope of the Uluguru Mountains

still be alive when we tried to catch them again 11 months later. When trying to retrieve the loggers in Kimboza Forest, we used a high number of mist nets (up to 35, 12.5 x 2 m nets) for 11 days, initially focusing on the part of the forest where the alethes were caught in 2023, but later we also set up nets in the more distant parts of the small forest. Loggers may have fallen off, although the harness we used is a well proven set-up, and perhaps the most likely explanation for our low recovery rate is that most of the birds have moved to other nearby lowland forests. Although we only succeeded in recapturing a single bird with logger, we believe the results of this study show that barometric

pressure loggers have the potential to reveal many new aspects of seasonal altitudinal migration.

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